

TEJASVINI SINGH

BTech - Bioengineering

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SUMMARY

I am a recent BTech Bioengineering graduate with a passion for integrating biology and engineering. My expertise includes bioinformatics, bio-fabrication, and biomedical image processing. I have practical experience through internships and projects. I aim to innovate and contribute to advancements in healthcare

EDUCATION

B-tech in Bioengineering

Mit-world peace university

09/2021 - 07/2025 Pune

Class 12 , HSC

Rao junior college of science

05/2020 - 07/2021 Thane

Professional Classical Dancer

Bhartiya sangeet samiti

01/2012 - Present Maharashtra

- Completed 9 professional exams

PROJECTS / INTERNSHIP

Report on Lung cancer Biomarkers

- Conducted an in-depth literature review on lung cancer biomarkers, focusing on genetic, protein, circulating, and imaging-based markers for early detection and personalized treatment strategies.
- Analyzed novel biomarkers such as TPX2 and ERCC1, and explored machine learning approaches in biomarker discovery and validation.
- Summarized clinical implications for improved prognosis and targeted therapies in lung cancer management.

Internship - Project Trainee

CSIR-National Chemical Laboratory

07/2024 - 12/2024 Pune

Polymer Science and Engineering Department under guidance of Dr. S Kiran

- Developed a biodegradable colon stent using PCL-grafted pectin for targeted drug delivery in colon cancer treatment
- Synthesized and characterized materials using techniques like NMR spectroscopy, DSC, FTIR
- Designed and optimized 3D stent models using Fusion 360 and printed it using a bio 3D printer
- Gained hands-on experience in 3D printing, polymer synthesis, and electrospinning

Silk Based Microspheres for Drug Delivery

- Prepared and optimized silk fibroin solutions at various concentrations; studied and analyzed properties of polyvinyl alcohol (PVA) to design suitable composite formulations.
- Developed composite solutions by blending silk fibroin with PVA in different ratios to enhance mechanical and film-forming properties.
- Fabricated and characterized both pure silk fibroin and silk fibroin-PVA composite films.

INTERESTS

Dancing

Photography

Organizing events

Sketching / painting

Volunteering

LANGUAGES

English

Hindi

Marathi

German (Basic)

CERTIFICATION

Molecular Docking (Discovery Boulevard)

Certification focused on the techniques of molecular docking

Molecular docking (NextGen Life Sciences)

Certification in molecular docking emphasizing practical applications

Biomedical imaging process using MATLAB

Course involving biomedical imaging processes using MATLAB

Vaccine technology (IIT Bombay)

Course on vaccine development provided by IIT Bombay

Biomedical nanotechnology from Nptel

Course on biomedical nanotechnology focused on recent advancements

KEY ACHIEVEMENTS

Best Poster Award

Presented original research poster titled "Biodegradable Stent for Colon Cancer" at BioConclave 2025, MIT-WPU; awarded **First Prize** by industry experts and faculty.

SKILLS

3D modeling & printing

Fusion 360, 3D printer operation (FDM), STL file preparation, Electrospinning

Programming

Python (Intermediate), R(Basic)

Software & tools

MATLAB, AutoCAD, SolidWorks

Bioinformatics Tools

PyMOL, AutoDock, Discovery Studio

Wet Lab Techniques

Polymer synthesis, Distillation

PCR, qPCR, Gel Electrophoresis, DNA/RNA Extraction

Analytical technique

Contact Angle Measurement, SEM

ADDITIONAL EXPERIENCE

Sponsorship Head

Syntropy - Annual Event conducted by Dept of Chemical Engineering

Led a team as Sponsorship Head, developing outreach strategies and negotiating with sponsors to secure funds for the event.

General secretary

Squad-Up Esports - Official Esports Club of MITWPU

Led the club's operations, organized events, and coordinated with members to enhance the club's performance.

Core member

TEDxMITWPU

Contributed to planning, organizing, and executing TEDx events.